

$$\text{PED}_{1-2-3_\alpha} = \frac{m_1 \boldsymbol{r}_{1_{\alpha 1-2 \perp 1-2-3 \parallel}}^2 + m_2 \boldsymbol{r}_{2_{\alpha 1-2 \perp 1-2-3 \parallel}}^2 + m_2 \boldsymbol{r}_{2_{\alpha 2-3 \perp 1-2-3 \parallel}}^2 + m_3 \boldsymbol{r}_{3_{\alpha 2-3 \perp 1-2-3 \parallel}}^2}{\sum m_j \boldsymbol{r}_{j_\alpha}^2} \quad (1)$$